# Exit Ticket · reflect and retrieve

## Exit Ticket

Splitting work means the components never need to communicate.

## Use key vocabulary from memory

1. Explain decomposition in your own words.

2. Explain top-down design in your own words.

3. Explain interface in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Lesson 066 · Problem Decomposition: Build a Useful Solution

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What makes a problem suitable for computation?

### Q2

What is an abstraction?

### Q3

What is a procedure?

### Q4

Stack initially empty: Push A, B, C; Pop; Push D. Give the stack top-to-bottom.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Define and predict — Decomposition divides a larger problem into manageable sub-problems whose inputs and outputs are clear.

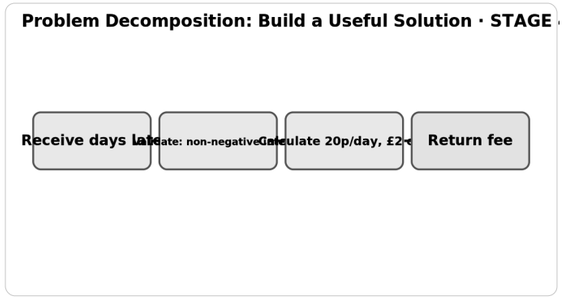
## Core idea 2

Explain — A top-down design begins with a main task, then refines it into independently testable functions or procedures.

## Core idea 3

Evaluate — Interfaces between subproblems matter: validation should happen before data are stored or a fee is charged.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Explain two benefits and one limitation of decomposition when a team builds a school booking app. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Identify three independent subtasks and their outputs for a library fine system. [6]

## Question 2

Write C# Fine(daysLate) for 20p/day capped at £2, zero for on-time returns. [6]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

decomposition — Split task into smaller problems.

top-down design — Start broad, refine into subtasks.

interface — Inputs/outputs between components.

modularity — Small cohesive program components.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

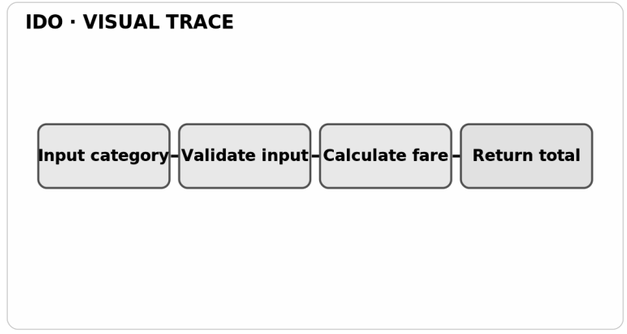
## Problem

Decompose a cinema price calculator for child/adult tickets.

## Follow the worked method

1. Follow the reasoning — State the interface for each subtask; every step has a specific responsibility.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Follow the reasoning
Reservation must occur before confirming success; check for failures.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.